

Amiet's leading- and trailing-edge solutions

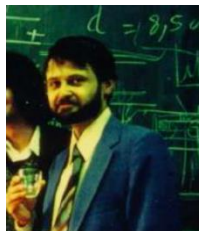
Overview

- **Amiet**¹⁻³ addressed the problems of **leading- and trailing-edge noise** by using the solution to the **Schwartzschild** problem, which was developed for electromagnetic wave scattering in the presence of a semi-infinite half-plane
 - Solution to Schwartzschild problem is derived using the **Wiener-Hopf method** (Sec. 20.2.2, Glegg & Devenport) and **only applies to a stationary surface in the absence of flow** (extension to uniform flow will follow)

Helmholtz equation

$$\frac{\partial^2 \hat{p}}{\partial x_1^2} + \frac{\partial^2 \hat{p}}{\partial x_2^2} + k^2 \hat{p} = 0$$

$$\text{BC: } \begin{cases} [\Delta \hat{p}_s(x_1)]_{x_1 \geq 0} = -P_o e^{ik_1 x_1} & \text{Kutta} \\ \left[\frac{\partial \hat{p}(x_1, 0)}{\partial x_2} \right]_{x_1 < 0} = 0 & \text{Impermeable} \end{cases}$$



Roy K. Amiet
July 19, 1942 –
Oct. 12, 2024



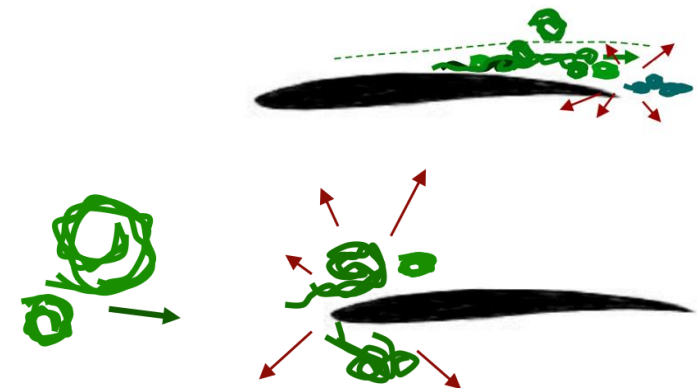
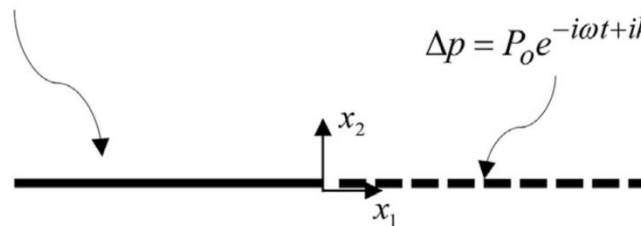
Karl Schwartzschild
Oct. 9, 1873 – May 11, 1916

Semi-infinite Flat Plate

$$\Delta p = P_o e^{-i\omega t + ik_1 x_1} + \Delta p_s \neq 0$$

Wake

$$\Delta p = P_o e^{-i\omega t + ik_1 x_1} + \Delta p_s = 0$$



¹ R. K. Amiet (1976) "Noise due to turbulent flow past a trailing edge", J. Sound Vib. 47, pp. 387–393

² R. K. Amiet (1978) "Effect of incident surface pressure field on noise due to turbulent flow past a trailing edge", J. Sound Vib. 57, pp. 305–306.

³ R. K. Amiet (1976) "High frequency thin airfoil theory for subsonic flow", AIAA J. 14, 1076–1082.

Amiet's leading- and trailing-edge solutions

Schwartzschild problem and TE solution

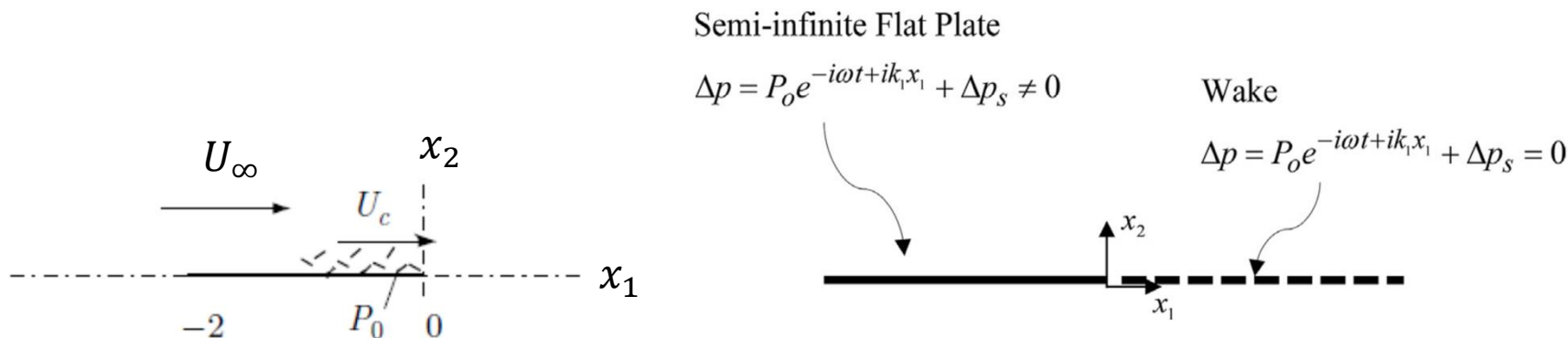
- Schwartzschild solved the problem to give the pressure jump over the plate ($x_1 < 0$):

$$\Delta p_s(x_1) = -\frac{2P_o}{\pi} \int_0^{\infty} \sqrt{\frac{|x_1|}{\xi}} \frac{e^{ik(\xi+|x_1|)+ik_1\xi}}{\xi + |x_1|} d\xi \quad x_1 < 0$$

- The generation of sound is interpreted as a linear wave diffraction mechanism

- Evaluating the integral above results in:

$$\Delta \hat{p}_s(x_1) = P_o e^{ik_1 x_1} [(1-i)E_2((k+k_1)|x_1|) - 1] \quad x_1 < 0$$

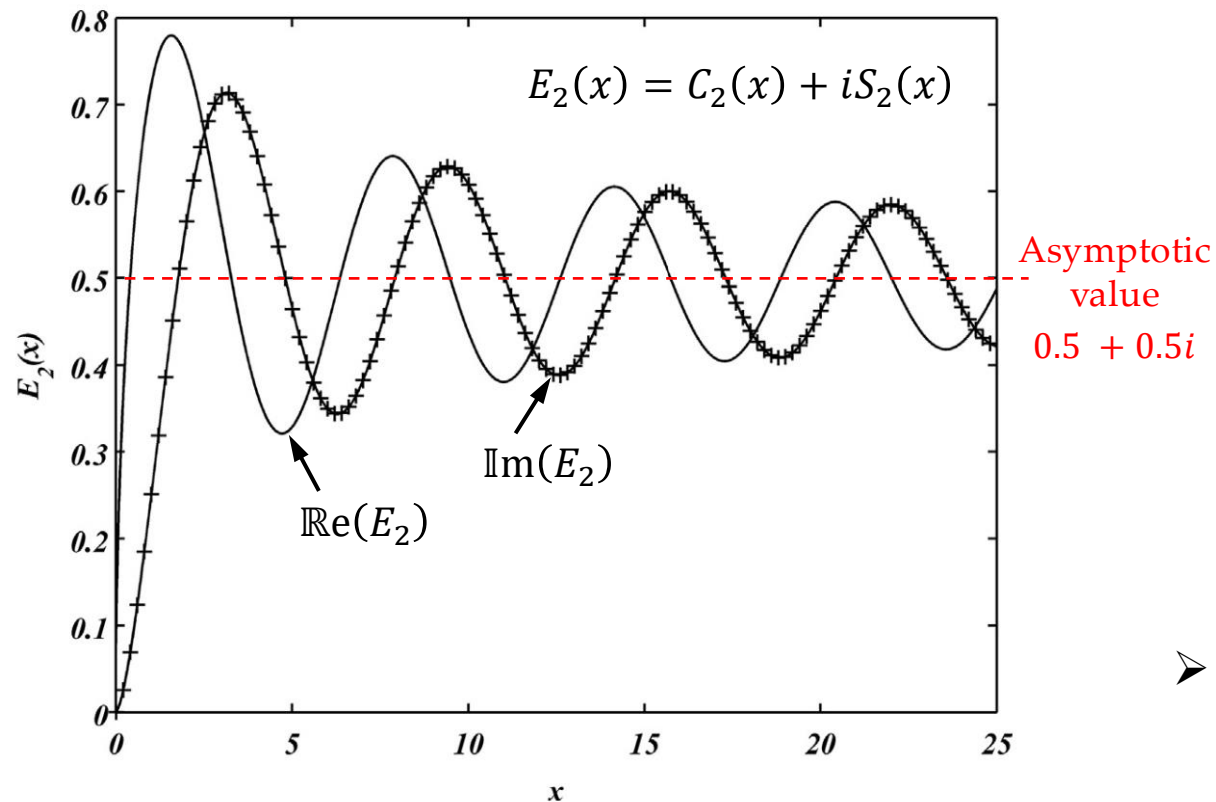


$E_2(x)$ - modified Fresnel integral

$$E_2(x) = \int_0^x \frac{e^{i\zeta}}{\sqrt{2\pi\zeta}} d\zeta$$

Amiet's leading- and trailing-edge solutions

Schwartzschild problem and TE solution



$E_2(x)$ - modified Fresnel integral

$$E_2(x) = \int_0^x \frac{e^{i\zeta}}{\sqrt{2\pi\zeta}} d\zeta$$

MATLAB implementation of $E_2(x)$:

`fresnelc(sqrt(2*x/pi)) + 1i*fresnels(sqrt(2*x/pi))`

- $E_2(x)$ appears often in the theory of edge scattering in several different forms

Amiet's leading- and trailing-edge solutions

Effect of uniform flow on Schwartzschild TE solution

- ❑ The Schwartzschild problem must be modified to account for the **mean flow over the surface**
- ❑ Also, since surface pressure perturbation can be three-dimensional, we extend our 2D analysis to 3D
- ❑ Thus, the pressure and velocity potential perturbations must satisfy the **convective wave eq.**:

$$\boxed{\frac{1}{c_\infty^2} \frac{D_\infty^2 p'}{Dt^2} - \nabla^2 p' = 0} \quad \text{where: } p' = -\rho_0 \frac{D_\infty \phi}{Dt}$$

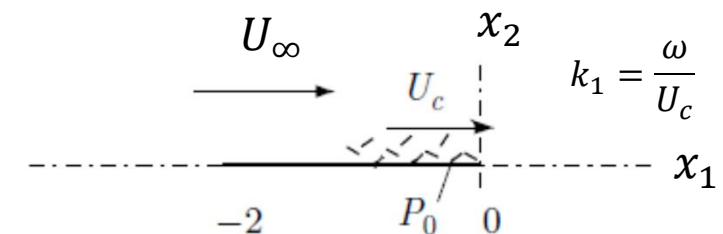
- Time derivative is expressed in the frame of reference of an observer who is moving relative to the medium
- ❑ Surface pressure perturbation $\Delta \hat{p}_s$ is considered harmonic, but is convected downstream in the direction of the flow, and can include a harmonic *spanwise variation*, so that the BC is modified to:

$$[\Delta \hat{p}_s(x_1, x_3)]_{x_1 \geq 0} = -P_0 e^{ik_1 x_1 + ik_3 x_3}$$

- ❑ Incident disturbance is harmonic in time and the surface geometry is independent of the spanwise direction, thus:

$$[\hat{p}(x_1, x_2, x_3)]_{3D} = [\hat{p}(x_1, x_2)]_{2D} e^{ik_3 x_3}$$

$$\Rightarrow p'(\mathbf{x}, t) = [\hat{p}(x_1, x_2)]_{2D} e^{-i\omega t + ik_3 x_3}$$



Amiet's leading- and trailing-edge solutions

Effect of uniform flow on Schwartzschild TE solution

- Substituting into the convective wave equation, we obtain:

$$\boxed{\beta^2 \frac{\partial^2 \hat{p}}{\partial x_1^2} + 2ikM \frac{\partial \hat{p}}{\partial x_1} + \frac{\partial^2 \hat{p}}{\partial x_2^2} + (k^2 - k_3^2)\hat{p} = 0} \quad k = \frac{\omega}{c_\infty} \quad \beta^2 = 1 - M^2 \quad M = \frac{U_\infty}{c_\infty}$$

- Solving the above results in the blade response to an incident pressure disturbance, which can later be used to evaluate trailing-edge noise:

$$\boxed{\Delta \hat{p}_s(x_1, x_3) = P_o g_{TE}(x_1, k_1, k_3) e^{ik_3 x_3}} \quad x_1 < 0 \quad A = 1 + \frac{\kappa + k_o M}{k_1}$$

$$\boxed{g_{TE}(x_1, k_1, k_3) = e^{ik_1 x_1} [(1 - i) E_2(k_1 A |x_1|) - 1]} \quad k_o = \frac{\omega}{\beta^2 c_\infty} \quad \kappa^2 = k_o^2 - \frac{k_3^2}{\beta^2}$$

- When $k_3 = 0$ and $U_c = U_\infty$: $A = \frac{1}{1 - M} > 0$

- Mean flow effect has a factor of $1/(1 - M)$, resulting in a response more rapid close to $x_1 = 0$

NOTE!

κ can be imaginary for large values of k_3 !
Yet, for $\Delta \hat{p}_s$ that couple to the acoustic far field κ must always be real!

Amiet's leading- and trailing-edge solutions

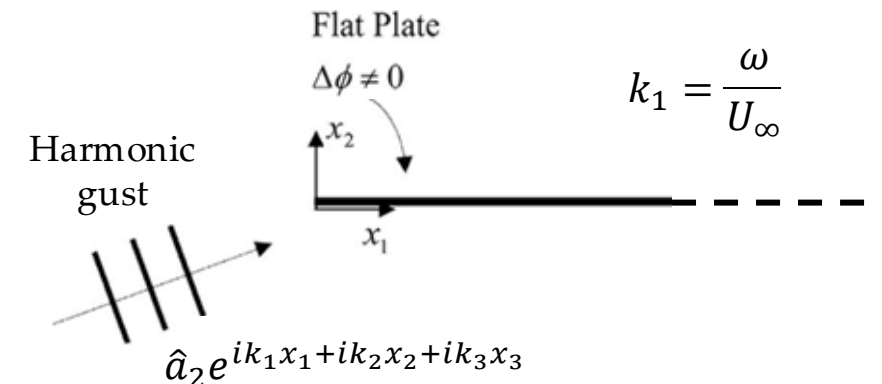
LE solution

- ❑ Effect of a **finite chord** is *essential* to solve the LE scattering problem due to an unsteady upwash gust
- ❑ Thus, we need to include the effect of both the LE and the TE in the solution, as follows:
 1. Solve the problem of an upwash gust incident on a semi-infinite flat plate
 2. Add a **TE correction** based on the Schwartzschild solution described previously to ensure that there is no pressure jump across the wake downstream of the blade TE
 - Correction introduces a small jump in potential upstream of the LE, so additional terms need to be added to correct this (this is a relatively small correction that can usually be ignored)
- ❑ The convective wave eq. is defined with the velocity potential perturbation :

$$\frac{1}{c_\infty^2} \frac{D_\infty^2 \phi}{Dt^2} - \nabla^2 \phi = 0$$

where: $\hat{\phi}(x_1, x_2, x_3)e^{-i\omega t} = \hat{\phi}(x_1, x_2)e^{-i\omega t + ik_3 x_3}$

$$\text{BC: } \begin{cases} [\Delta \hat{\phi}(x_1, x_3)]_{x_1 < 0} = 0 \\ \left[\frac{\partial \hat{\phi}(x_1, 0, x_3)}{\partial x_2} + \hat{a}_2 e^{ik_1 x_1 + ik_3 x_3} \right]_{x_1 > 0} = 0 \end{cases}$$



Amiet's leading- and trailing-edge solutions

LE solution

- Solving the LE problem results in the **semi-infinite plate response** to an unsteady upwash gust:

$$\Delta \hat{p}^{(1)}(x_1, x_3) = \pi \rho_\infty \hat{a}_2 U_\infty g^{(1)}(x_1, k_3, \omega, M) e^{ik_3 x_3}$$

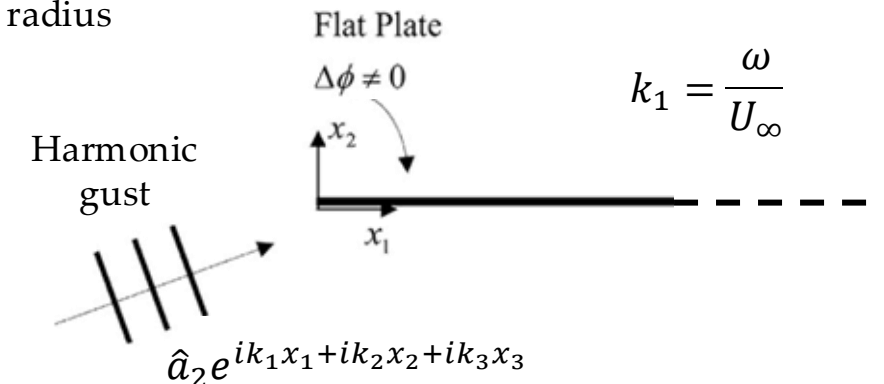
$$\beta^2 = 1 - M^2 \quad M = \frac{U_\infty}{c_\infty}$$

$$g^{(1)}(x_1, k_3, \omega, M) = \frac{-2e^{i(\kappa - Mk_0)x_1 + i\pi/4}}{\pi^{3/2}(k_1 + \kappa\beta^2)^{1/2}x_1^{1/2}}$$

$$x_1 > 0 \quad k_0 = \frac{\omega}{\beta^2 c_\infty} \quad \kappa^2 = k_0^2 - \frac{k_3^2}{\beta^2}$$

$$k_1 + k_0 M = \frac{k_1}{\beta^2}$$

- Notice the pressure jump is singular at the LE ($x_1 = 0$) and tends to zero at large distances ($x_1 \rightarrow \infty$)
- Due to thin airfoil theory...
 - In reality, the velocity at the LE will be finite due to viscous effects and LE radius



Amiet's leading- and trailing-edge solutions

LE solution

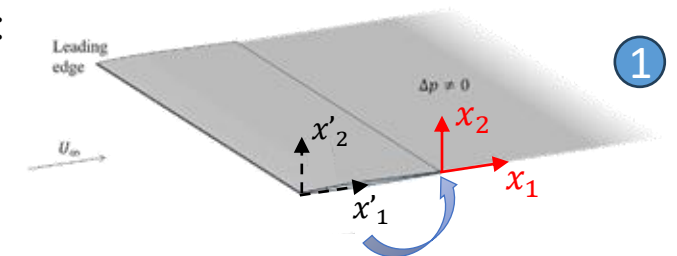
- ❑ For a finite chord blade, results need to be **corrected** for the **TE effect** and the **Kutta condition**
- ❑ **Correction to the previous semi-infinite LE result is done via TE scattering theory** (*Schwartzschild...*)
 - Yet, the TE result was for a semi-infinite flat plate, and so there will be a residual error that occurs at the LE
- ❑ While successive corrections can be done, we shall consider only a **first-order TE correction** (*sufficient*) obtained by solving the TE problem with the incident $\Delta\hat{p}$ obtained from the LE solution

- ❑ First, **rewrite the solution of the LE problem with the origin at the TE**:

$$\Delta\hat{p}^{(1)}(x_1, x_3) = \pi\rho_\infty\hat{a}_2U_\infty \left\{ \frac{-2e^{i(\kappa-Mk_0)(x_1+c)+i\pi/4}}{\pi(k_1 + \kappa\beta^2)^{1/2}\sqrt{\pi(x_1+c)}} \right\} e^{ik_3x_3}$$

- ❑ Perform the full Wiener–Hopf procedure on the function is *difficult*
- ❑ Thus, **a simplified result is obtained in the high frequency limit**, where **amplitude variation with x_1 is ignored** ($\sqrt{x_1+c} \approx \sqrt{c}$), resulting in the following **approximation of the pressure near the TE**:

$$\Delta\hat{p}^{(1)}(x_1, x_3) \approx \pi\rho_\infty\hat{a}_2U_\infty \left\{ \frac{-2e^{i(\kappa-Mk_0)c+i\pi/4}}{\pi(k_1 + \kappa\beta^2)^{1/2}\sqrt{\pi c}} \right\} e^{i(\kappa-Mk_0)x_1+ik_3x_3}$$



Phase variation along the plate is *unaltered* (important for scattering!), but **x_1 -dependent amplitude** $1/\sqrt{x_1+c}$ is replaced by its value at the TE, $1/\sqrt{c}$

Amiet's leading- and trailing-edge solutions

LE solution

- We can now **perform the TE analysis** (as previously performed using the Wiener–Hopf procedure), to find the TE correction as the additional pressure jump:

$$\Delta \hat{p}^{(2)}(x_1, x_3) = \frac{-2\pi\rho_\infty \hat{a}_2 U_\infty e^{i(\kappa - Mk_0)(x_1 + c) + i\pi/4}}{\pi(k_1 + \kappa\beta^2)^{1/2} \sqrt{\pi c}} [(1 - i)E_2(2\kappa|x_1|) - 1] e^{ik_3 x_3}$$

Modified Fresnel integral

$$E_2(x) = \int_0^x \frac{e^{i\zeta}}{\sqrt{2\pi\zeta}} d\zeta$$

- We relocate the origin of the coordinate system to the LE, obtaining the corrected non-dimensional surface pressure as:

$$\Delta \hat{p}^{(1+2)} = \Delta \hat{p}_s(x_1, x_3) = \pi\rho_\infty \hat{a}_2 U_\infty g_{LE}(x_1, k_3, \omega, M) e^{ik_3 x_3}$$

$$g_{LE}(x_1, k_3, \omega, M) = \frac{-2e^{i(\kappa - Mk_0)x_1 + i\pi/4}}{\pi(k_1 + \kappa\beta^2)^{1/2} \sqrt{\pi c}} \left[\sqrt{\frac{c}{x_1}} + (1 - i)E_2(2\kappa|x_1 - c|) - 1 \right] \quad 0 < x_1 < c$$

Although the above is just a **first-order approximation**, it accounts for the **major TE effects**, ensuring that the Kutta condition is satisfied

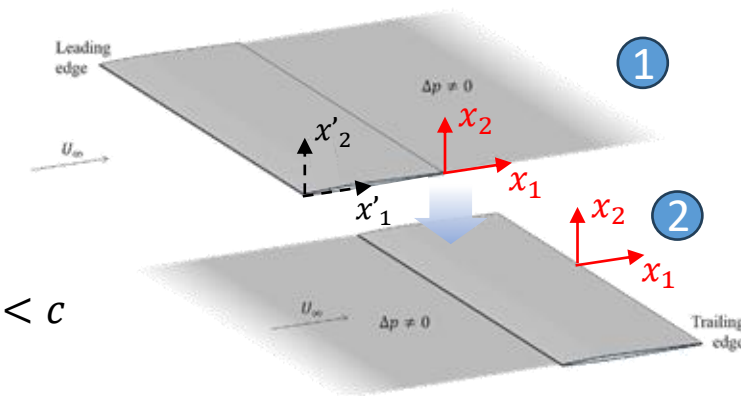
When $c \gg \lambda$ ($\kappa c \rightarrow \infty$):

$$\Rightarrow E_2(2\kappa c) = 0.5 + 0.5i$$

$$(1 - i)E_2(2\kappa c) - 1$$

$$\approx 0 + \mathcal{O}\left(\frac{1}{\sqrt{2\kappa c}}\right)$$

Aircraft Aeroacoustics $\Rightarrow$ small correction for high freq.



Flat Plate

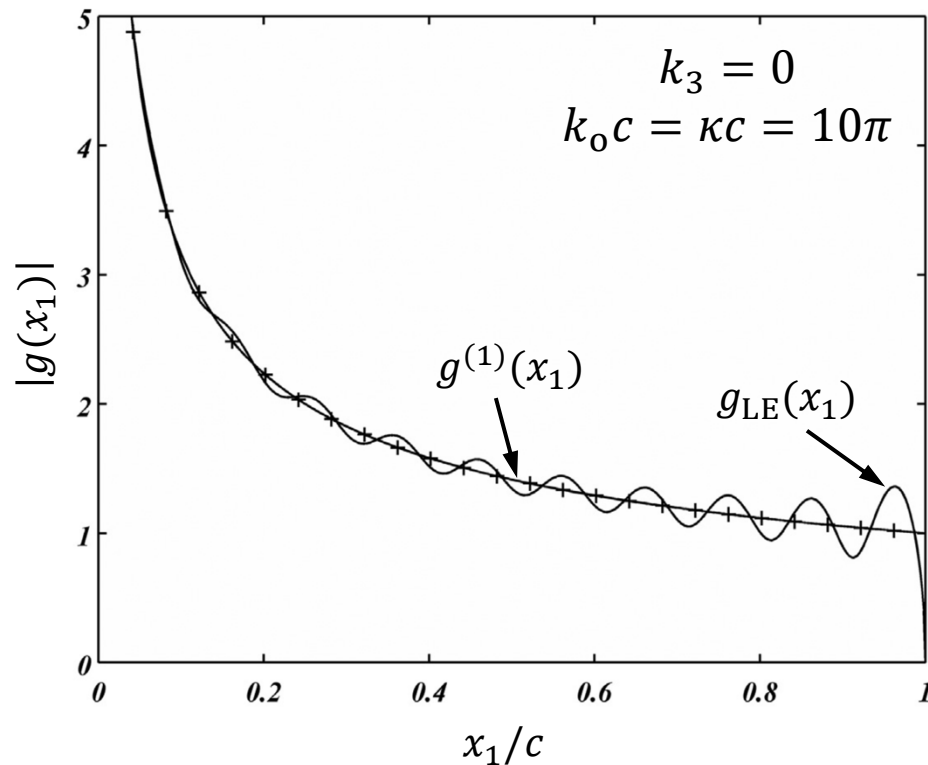
$$\Delta \phi \neq 0$$

Harmonic gust

$$\hat{a}_2 e^{ik \cdot x}$$

Amiet's leading- and trailing-edge solutions

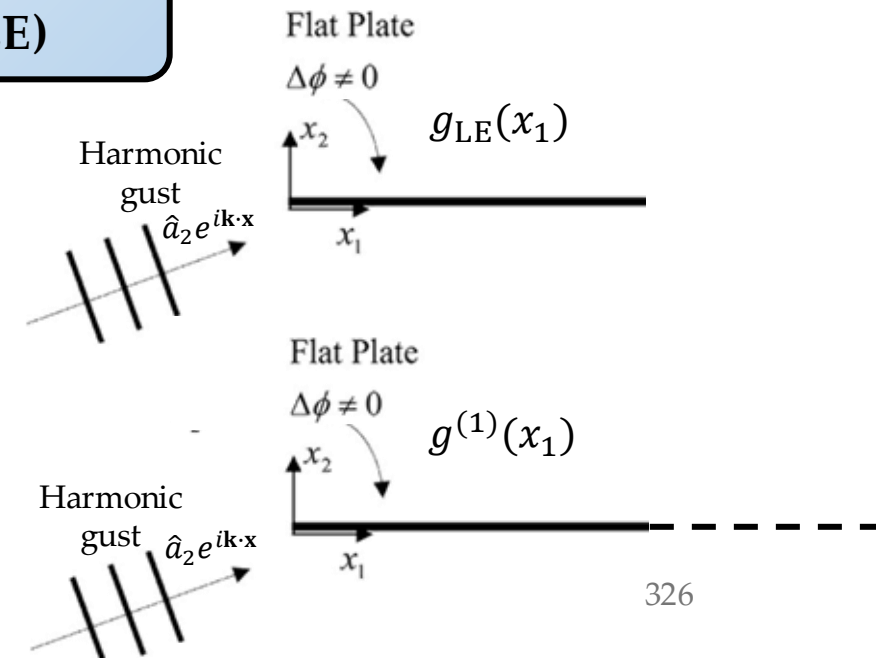
LE solution



$$g_{LE}(x_1, k_3, \omega, M) = \frac{-2e^{i(\kappa - Mk_0)x_1 + i\pi/4}}{\pi(k_1 + \kappa\beta^2)^{1/2}\sqrt{\pi c}} \left[\sqrt{\frac{c}{x_1}} + (1-i)E_2(2\kappa|x_1 - c|) - 1 \right] \quad 0 < x_1 < c$$

$$g^{(1)}(x_1, k_3, \omega, M) = \frac{-2e^{i(\kappa - Mk_0)x_1 + i\pi/4}}{\pi^{3/2}(k_1 + \kappa\beta^2)^{1/2}x_1^{1/2}} \quad x_1 > 0$$

TE correction effect diminished upstream, as $x_1 \rightarrow 0$ (LE)



Amiet's leading- and trailing-edge solutions

LE solution for an arbitrary gust

- So far, we analyzed a harmonic gust of the form $\hat{a}_2 e^{ik_1(x_1 - U_\infty t) + ik_3 x_3}$ in the plane of the blade ($x_2 = 0$)
- To extend these results to an **arbitrary gust**, we consider the upwash velocity on the blade surface $x_2 = 0$ to be the **superposition of gusts with a spectrum of wavenumbers**:

$$u_2(x_1 - U_\infty t, 0, x_3) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \tilde{u}_2(\mathbf{k}) e^{ik_1(x_1 - U_\infty t) + ik_3 x_3} d\mathbf{k}$$

- The **pressure jump** across the blade is then obtained by **superimposing the contributions from each wavenumber component** of the gust, so using the LE solution and *substituting* $\tilde{u}_2(\mathbf{k})$ for $\hat{a}_2$:

$$\Delta p(x_1, x_3, t) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \pi \rho_\infty U_\infty \tilde{u}_2(\mathbf{k}) g_{\text{LE}}(x_1, k_3, \omega, M) e^{-i\omega t + ik_3 x_3} d\mathbf{k} \quad k_1 = \frac{\omega}{U_\infty}$$

- Integrand over k_1 is simply an inverse Fourier transform with respect to time ($d\omega = U_\infty dk_1$), so:

$$p(\mathbf{x}, t) = \int_{-\infty}^{\infty} \tilde{p}(\mathbf{x}, \omega) e^{-i\omega t} d\omega \quad \Rightarrow \quad \Delta \tilde{p}(x_1, x_3, \omega) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \pi \rho_\infty \tilde{u}_2(\omega/U_\infty, k_2, k_3) g_{\text{LE}}(x_1, k_3, \omega, M) e^{ik_3 x_3} dk_2 dk_3$$

$$\Delta \hat{p}_s(x_1, x_3) = \pi \rho_\infty \hat{a}_2 U_\infty g_{\text{LE}}(x_1, k_3, \omega, M) e^{ik_3 x_3}$$

Amiet's leading- and trailing-edge solutions

LE solution for an arbitrary gust

- We can further simplify the previous equation by assuming the **gust response is independent of k_2**
 - Valid assumption for a flat plate at $x_2 = 0$ plane



$$\Delta \tilde{p}(x_1, x_3, \omega) = \int_{-\infty}^{\infty} \pi \rho_{\infty} \tilde{\tilde{w}}_2(k_1, k_3) g_{LE}(x_1, k_3, \omega, M) e^{ik_3 x_3} dk_3$$

$$k_1 = \frac{\omega}{U_{\infty}}$$

Planar wavenumber transform
of $u_2(x_1, 0, x_3)$ in the plane $x_2 = 0$

$$\tilde{\tilde{w}}_2(k_1, k_3) = \int_{-\infty}^{\infty} \tilde{\tilde{u}}_2(k_1, k_2, k_3) dk_2 = \frac{1}{(2\pi)^2} \int_{-R_{\infty}}^{R_{\infty}} \int_{-R_{\infty}}^{R_{\infty}} u_2(x_1, 0, x_3) e^{-ik_1 x_1 - ik_3 x_3} dx_1 dx_3$$

Amiet's leading- and trailing-edge solutions

Acoustic far-field noise

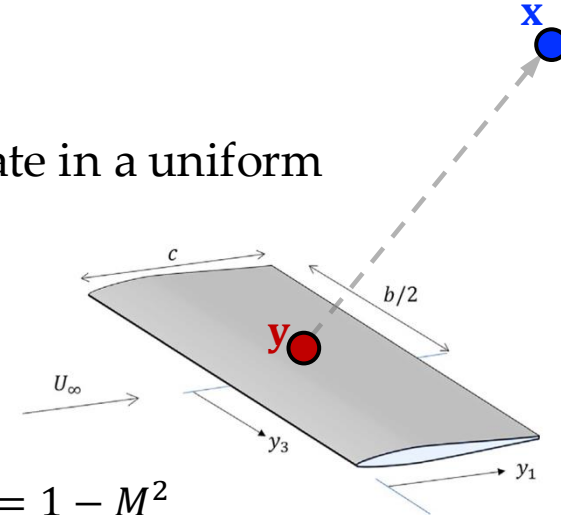
- The **acoustic far-field noise** ($|\mathbf{x}| \gg |\mathbf{y}|$) can be computed for a stationary flat plate in a uniform flow by using the **Prandtl-Glauert transformation**:

$$(\tilde{p}(\mathbf{x}, \omega) c_\infty^2)_{\text{dipole}} \approx \frac{-i\omega x_2 e^{ik|\mathbf{x}|}}{4\pi c_\infty |\mathbf{x}|^2} \int_0^c \int_{-b/2}^{b/2} \Delta \tilde{p}(\mathbf{y}, \omega) e^{-ik_1 x_1 y_1 / |\mathbf{x}| - ik_3 x_3 y_3 / |\mathbf{x}|} dy_1 dy_3$$

Prandtl-Glauert transformation

Wave field in a moving frame is the same as a wave field in a stationary frame with modified space/time variables

$$\xi = (x_1, \beta x_2, \beta x_3) \quad \text{and} \quad t_g = t + \frac{M x_1}{c_\infty \beta^2}$$



$$\beta^2 = 1 - M^2$$

$$k_0 = \frac{\omega}{\beta^2 c_\infty}$$

$$r_g = \sqrt{x_1^2 + \beta^2(x_2^2 + x_3^2)}$$

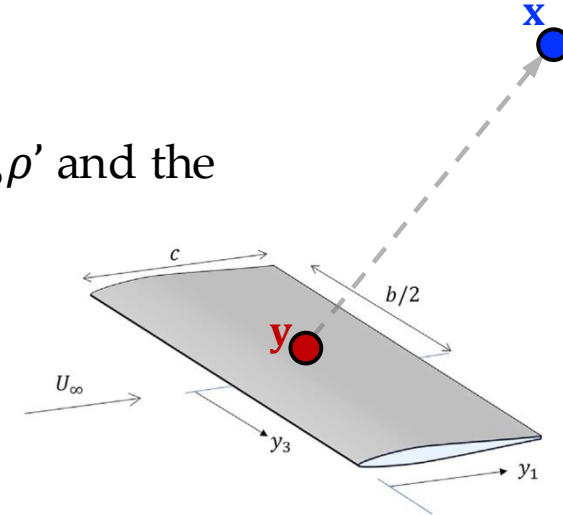
$$(\tilde{p}(\mathbf{x}, \omega) c_\infty^2)_{\text{dipole}} \approx \frac{-i\omega x_2 e^{ik_0 r_g}}{4\pi c_\infty r_g^2} \int_{-b/2}^{b/2} \int_0^c \Delta \tilde{p}(\mathbf{y}, \omega) e^{-ik_0 x_1 y_1 / r_g - ik_0 x_3 y_3 \beta^2 / r_g - ik_0 M(x_1 - y_1)} dy_1 dy_3$$

Amiet's leading- and trailing-edge solutions

Acoustic far-field noise

- We can express the result in terms of the acoustic pressure perturbation $p' = c_\infty^2 \rho'$ and the **wavenumber transform of the surface pressure**, so:

$$\tilde{p}(\mathbf{x}, \omega) \approx \left(\frac{-i\pi\omega x_2 e^{ik_0 r_g - ik_0 M x_1}}{c_\infty r_g^2} \right) \Delta \tilde{p}(k_1^{(o)}, k_3^{(o)}, \omega)$$



2D wavenumber frequency transform
(chordwise-spanwise)
of the pressure jump over the airfoil planform

$$\Delta \tilde{p}(k_1^{(o)}, k_3^{(o)}, \omega) = \frac{1}{(2\pi)^2} \int_{-b/2}^{b/2} \int_0^c \Delta \tilde{p}(\mathbf{y}, \omega) e^{-ik_1^{(o)} y_1 - ik_3^{(o)} y_3} dy_1 dy_3$$

Acoustic wavenumbers
in the observer direction

$$k_1^{(o)} = k_0 \left(\frac{x_1}{r_g} - M \right)$$

$$k_3^{(o)} = \frac{k_0 x_3 \beta^2}{r_g}$$

$$\beta^2 = 1 - M^2$$

$$k_0 = \frac{\omega}{\beta^2 c_\infty}$$

$$r_g = \sqrt{x_1^2 + \beta^2(x_2^2 + x_3^2)}$$

Amiet's leading- and trailing-edge solutions

Acoustic far-field LE noise

- The 2D wavenumber frequency transform of the surface pressure $\Delta\tilde{p}(k_1^{(o)}, k_3^{(o)}, \omega)$ is computed by using **Amiet's LE solution for arbitrary gust**:

substitute into

$$\Delta\tilde{p}(\mathbf{y}_1, \mathbf{y}_3, \omega) = \int_{-\infty}^{\infty} \pi \rho_{\infty} \tilde{w}_2(k_1, k_3) g_{LE}(\mathbf{y}_1, k_3, \omega, M) e^{ik_3 \mathbf{y}_3} dk_3$$

$$\Delta\tilde{p}(k_1^{(o)}, k_3^{(o)}, \omega) = \frac{1}{(2\pi)^2} \int_{-b/2}^{b/2} \int_0^c \Delta\tilde{p}(\mathbf{y}, \omega) e^{-ik_1^{(o)} \mathbf{y}_1 - ik_3^{(o)} \mathbf{y}_3} dy_1 dy_3$$

Inverse Fourier transform
over k_3 + forward
transform over y_3

$$\Delta\tilde{p}(k_1^{(o)}, k_3^{(o)}, \omega) = \frac{1}{2} \rho_{\infty} c \tilde{w}_2(\omega/U_{\infty}, k_3^{(o)}) \mathcal{L}_{LE}(k_1^{(o)}, k_3^{(o)}, \omega, M)$$

**Unsteady lift
transfer function**

$$\mathcal{L}_{LE}(k_1^{(o)}, k_3^{(o)}, \omega, M) = \frac{1}{c} \int_0^c g_{LE}(\mathbf{y}_1, k_3^{(o)}, \omega, M) e^{-ik_1^{(o)} \mathbf{y}_1} dy_1$$

**Far-field
LE noise**

$$\tilde{p}(\mathbf{x}, \omega) \approx \left(\frac{i\pi\omega x_2 e^{ik_0 r_g - ik_0 M x_1}}{c_{\infty} r_g^2} \right) \frac{1}{2} \rho_{\infty} c \tilde{w}_2(\omega/U_{\infty}, k_3^{(o)}) \mathcal{L}_{LE}(k_1^{(o)}, k_3^{(o)}, \omega, M)$$

$$\beta^2 = 1 - M^2$$

$$k_0 = \frac{\omega}{\beta^2 c_{\infty}}$$

$$r_g = \sqrt{x_1^2 + \beta^2(x_2^2 + x_3^2)}$$

$$k_1^{(o)} = k_0 \left(\frac{x_1}{r_g} - M \right)$$

$$k_3^{(o)} = \frac{k_0 x_3 \beta^2}{r_g}$$

Amiet's leading- and trailing-edge solutions

Acoustic far-field TE noise

- The full wavenumber frequency transform of the surface pressure $\Delta\tilde{p}(k_1^{(o)}, k_3^{(o)}, \omega)$ is related to the spanwise wavenumber transform as:

$$\Delta\tilde{p}(k_1^{(o)}, k_3^{(o)}, \omega) = \frac{1}{2\pi} \int_{-c}^0 \Delta\tilde{p}(\mathbf{y}_1, k_3^{(o)}, \omega) e^{-ik_1^{(o)}\mathbf{y}_1} d\mathbf{y}_1$$



Far-field
TE noise

$$\tilde{p}(\mathbf{x}, \omega) \approx \frac{-i\omega x_2 e^{ik_0 r_g - ik_0 M x_1}}{2c_\infty r_g^2} \int_{-c}^0 \Delta\tilde{p}(\mathbf{y}_1, k_3^{(o)}, \omega) e^{-ik_1^{(o)}\mathbf{y}_1} d\mathbf{y}_1$$

$$\beta^2 = 1 - M^2$$

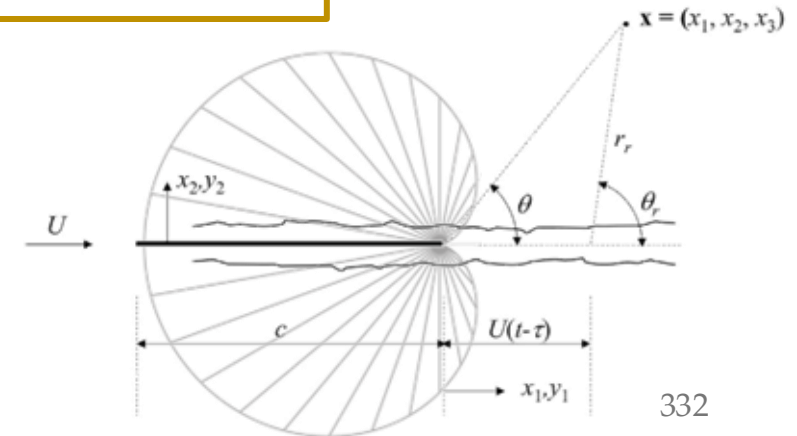
$$k_0 = \frac{\omega}{\beta^2 c_\infty}$$

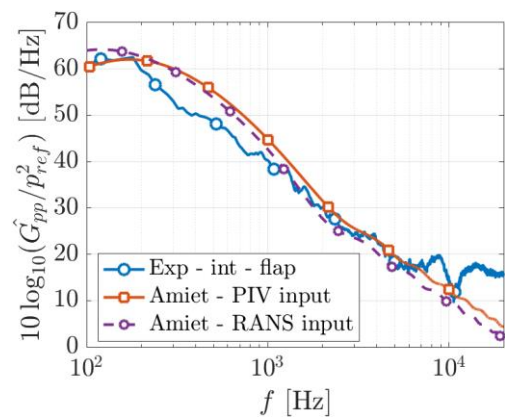
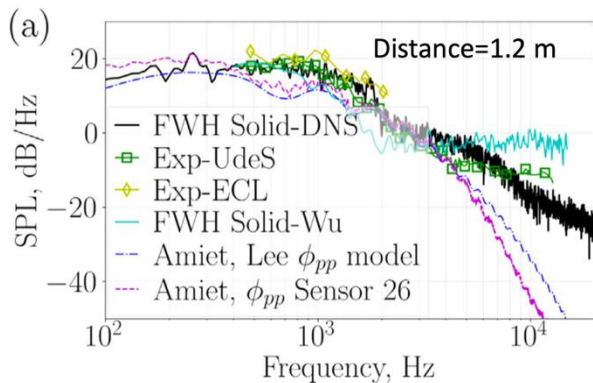
$$r_g = \sqrt{x_1^2 + \beta^2(x_2^2 + x_3^2)}$$

$$k_1^{(o)} = k_0 \left(\frac{x_1}{r_g} - M \right)$$

$$k_3^{(o)} = \frac{k_0 x_3 \beta^2}{r_g}$$

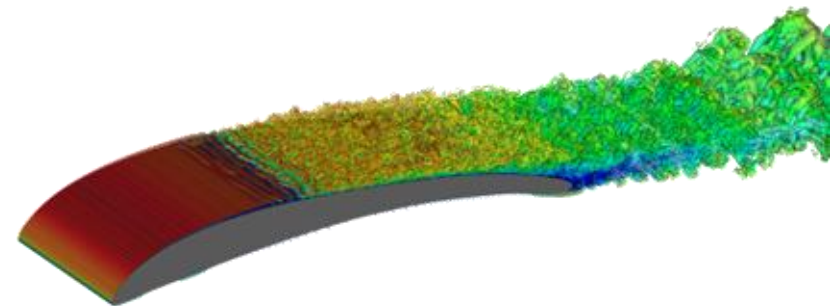
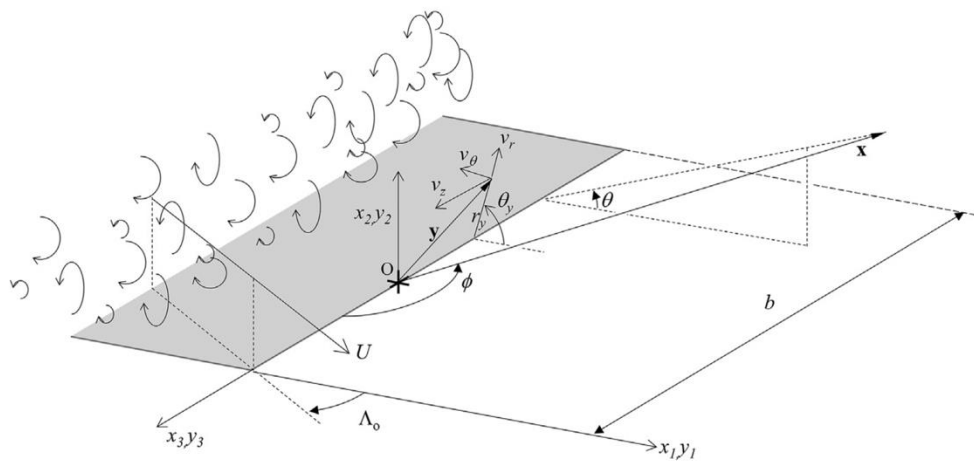
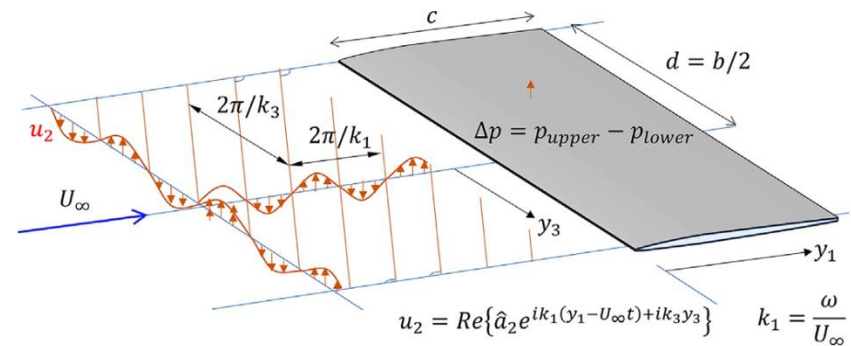
Statistically *identical* turbulent
BLs are assumed to form on
either side of the plate



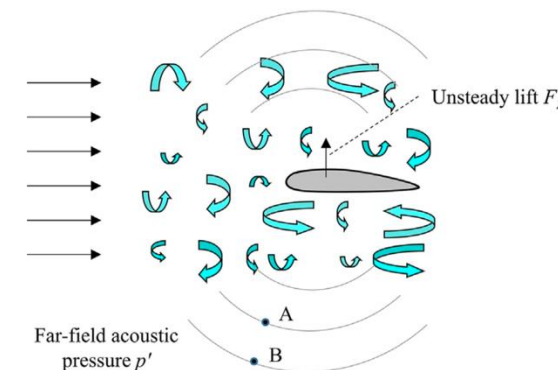


Chapter 5

Airfoil Self-Noise



Time correlations and frequency spectra



- Recall the definition of the **auto-spectral density** of p' :

$$S_{pp}(\omega) = \frac{1}{2\pi} \int_{-T}^T R_{pp}(\tau) e^{i\omega\tau} d\tau$$

$$R_{pp}(\tau) = \int_{-\infty}^{+\infty} S_{pp}(\omega) e^{-i\omega\tau} d\omega$$

inverse

- $R_{pp}(\tau)$ is the **time-delay auto-correlation function**

- Average of the signal multiplied by itself at a later time

$$R_{pp}(\tau) = E[p(t)p(t + \tau)]$$

- **E – expected value** – mean value of a stochastic variable over many realizations N

$$E[p(t)] = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N p_n(t)$$

- We can average the time delay correlation $R_{pp}(\tau)$ over time without changing its value:

$$R_{pp}(\tau) = \frac{1}{2T} \int_{-T}^T E[p(t)p(t + \tau)] dt$$

Substituting to
 S_{pp} definition

$$S_{pp}(\omega) = \frac{1}{4\pi T} E \left[\int_{-T}^T \int_{-T}^T p(t)p(t + \tau) e^{i\omega\tau} dt d\tau \right]$$

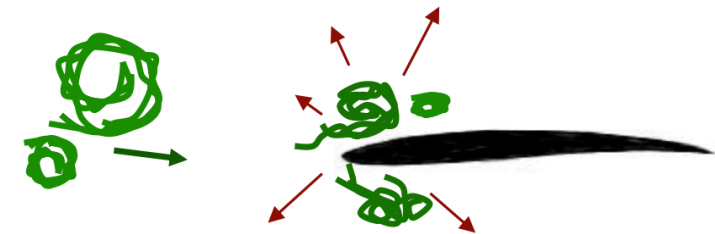
For $\omega \gg \pi/T$:

$$S_{pp}(\omega) = \frac{\pi}{T} E[\tilde{p}^*(\omega)\tilde{p}(\omega)]$$

$\tilde{p}^*$ – conjugate

Amiet's LE noise model

An airfoil in a turbulent stream



- We previously found the far-field acoustic pressure of a blade of large span in a turbulent flow:

$$\tilde{p}(\mathbf{x}, \omega) \approx \left(\frac{i\pi\omega x_2 e^{ik_0 r_g - ik_0 M x_1}}{c_\infty r_g^2} \right) \frac{1}{2} \rho_\infty c \tilde{W}_2(\omega/U_\infty, k_3^{(o)}) \mathcal{L}_{LE}(k_1^{(o)}, k_3^{(o)}, \omega, M)$$

- Far-field spectral density of the acoustic pressure:

$$S_{pp}(\omega) = \frac{\pi}{T} E[\tilde{p}^*(\omega)\tilde{p}(\omega)] \quad \Rightarrow \quad S_{pp}(\mathbf{x}, \omega) \approx \left(\frac{\pi\omega x_2}{c_\infty r_g^2} \right)^2 \left| \frac{1}{2} \rho_\infty c \mathcal{L}_{LE}(k_1^{(o)}, k_3^{(o)}, \omega, M) \right|^2 \underbrace{\frac{\pi}{T} E[|\tilde{W}_2(\omega/U_\infty, k_3^{(o)})|^2]}_{= \frac{R_\infty^2}{\pi T} \phi_{22}(\omega/U_\infty, k_3^{(o)}) = \frac{U_\infty b}{2\pi} \phi_{22}(\omega/U_\infty, k_3^{(o)})}$$

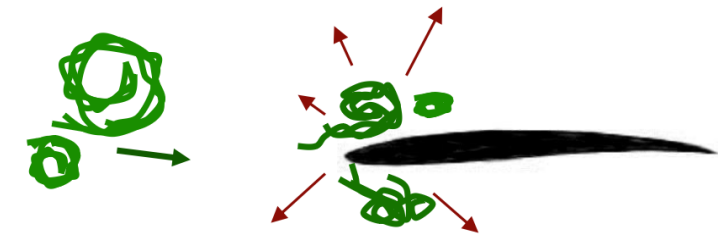
ϕ_{22} – 2D (planar) wavenumber spectra of normal velocity fluctuations in x_1 and x_3

$T = R_\infty/U_\infty$ – averaging time – the time for the turbulent volume to pass over the blade

$R_\infty = b/2$ – turbulent region size – taken as the wetted span of the blade

Amiet's LE noise model

An airfoil in a turbulent stream



$$S_{pp}(\mathbf{x}, \omega) \approx \left(\frac{\rho_{\infty} k c \mathbf{x}_2}{2r_g^2} \right)^2 \left(\frac{\pi U_{\infty} b}{2} \right) \phi_{22} \left(\omega / U_{\infty}, k_3^{(o)} \right) \left| \mathcal{L}_{LE} \left(k_1^{(o)}, k_3^{(o)}, \omega, M \right) \right|^2$$

directivity
coefficient

×

Source term
spectra

×

Unsteady lift
response function

- von-Karman
- Liepmann

- c – airfoil chord
- b – span
- r_g – effective observer distance
- $U_{\infty}, \rho_{\infty}, c_{\infty}$ – freestream velocity, density, and speed of sound
- M – freestream Mach number, U_{∞}/c_{∞}
- ω – angular frequency [rad/sec]
- k – wavenumber [rad/m]
- ϕ_{22} – 2D (planar) wavenumber spectra of normal velocity fluctuations in x_1 and x_3
- $\mathcal{L}_{LE}$ – Unsteady lift transfer function

$$k = \frac{\omega}{c_{\infty}} \quad k_1 = \frac{\omega}{U_{\infty}}$$

$$k_o = \frac{\omega}{\beta^2 c_{\infty}} \quad \beta^2 = 1 - M^2$$

$$r_g = \sqrt{\mathbf{x}_1^2 + \beta^2(\mathbf{x}_2^2 + \mathbf{x}_3^2)}$$

$$k_1^{(o)} = k_o \left(\frac{\mathbf{x}_1}{r_g} - M \right)$$

$$k_3^{(o)} = \frac{k_o \mathbf{x}_3 \beta^2}{r_g}$$

Amiet's LE noise model

An airfoil in a turbulent stream

□ High-frequency acoustic loading: $\mu_a = \frac{k_1 c M}{2\beta^2} > \frac{\pi}{4}$

➤ We split the unsteady loading term into two contributions:

$$\mathcal{L}_{LE} = \mathcal{L}_1 + \mathcal{L}_2$$

$$\mathcal{L}_1(k_1, x_1) = \frac{\sqrt{2}}{\pi\beta\sqrt{\mu_h(1+M)}\Theta_1} E_2^*(2\Theta_1) e^{i\Theta_2}$$

Main LE scattering
a semi-infinite plate

Back-scattering
edge correction

$$\mathcal{L}_1(k_1, x_1) = \frac{e^{i\Theta_2} \left\{ i(1 - e^{-2i\Theta_1}) + (1 - i) \left[E_2^*(4\mu_a\beta^2) - \sqrt{\frac{2}{1+x_1/r_g}} E_2^*(2\Theta_3) e^{-2i\Theta_1} \right] \right\}}{\pi\Theta_1\beta\sqrt{2\pi\mu_h(1+M)}}$$

$$\mu_h = \frac{k_1 c}{2\beta^2}$$

$$\Theta_1 = \mu_a \left(1 - \frac{x_1}{r_g} \right)$$

$$\Theta_2 = \beta^2 \mu_h + \mu_a \left(M - \frac{x_1}{r_g} \right) - \frac{\pi}{4}$$

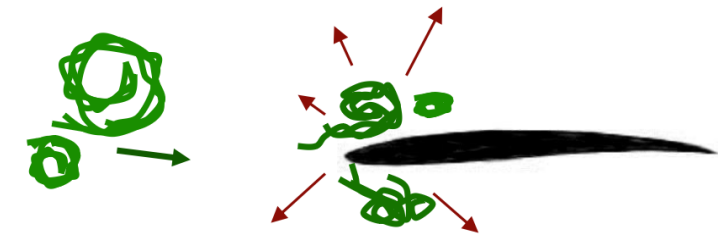
$$\Theta_3 = \mu_a \left(1 + \frac{x_1}{r_g} \right)$$

MATLAB implementation of $E_2^*(x)$:

`fresnelc(sqrt(2*x/pi)) - 1i*fresnels(sqrt(2*x/pi))`

$E_2^*(x)$ - **conjugate**
Fresnel integral

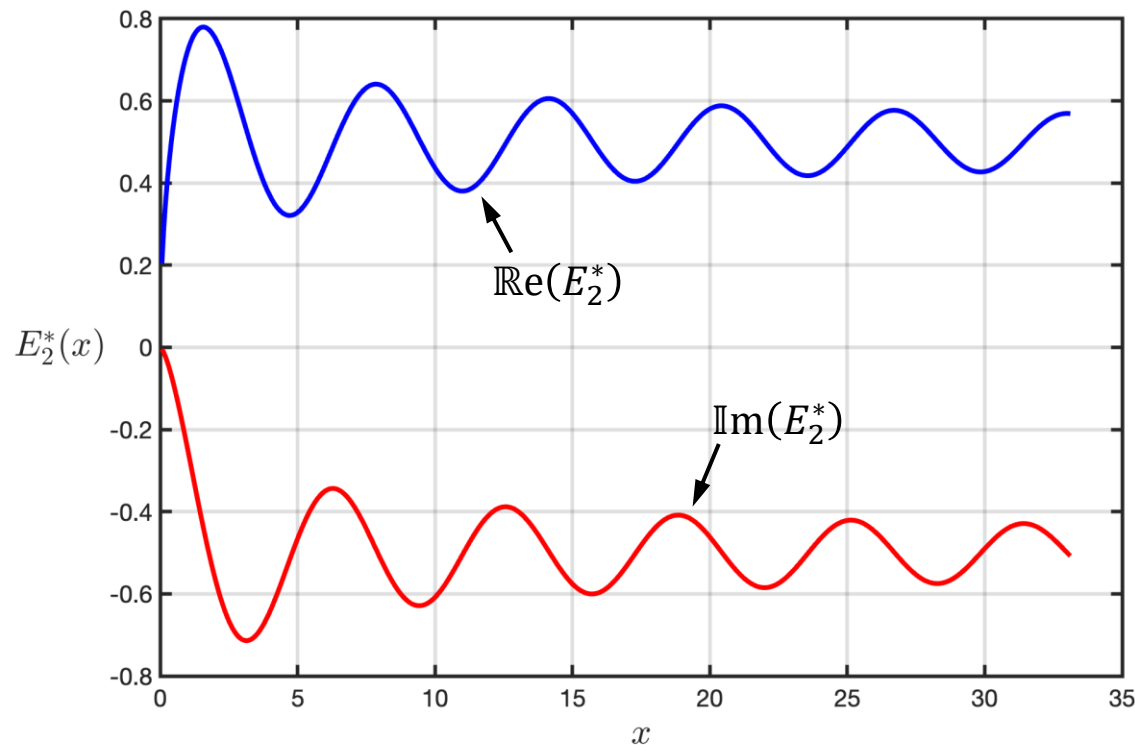
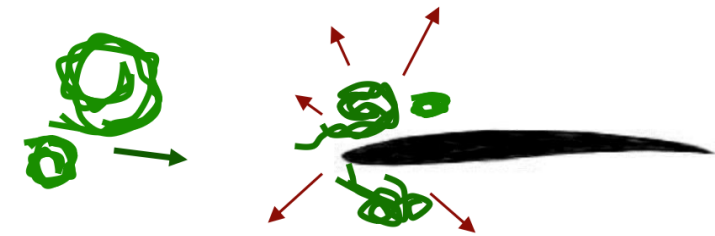
$$E_2^*(x) = \int_0^x \frac{e^{-i\zeta}}{\sqrt{2\pi\zeta}} d\zeta$$



Amiet's LE noise model

An airfoil in a turbulent stream

□ High-frequency acoustic loading: $\mu_a = \frac{k_1 c M}{2\beta^2} > \frac{\pi}{4}$



$E_2^*(x)$ - conjugate
Fresnel integral

$$E_2^*(x) = \int_0^x \frac{e^{-i\zeta}}{\sqrt{2\pi\zeta}} d\zeta$$

MATLAB implementation of $E_2^*(x)$:
`fresnelc(sqrt(2*x/pi)) - 1i*fresnels(sqrt(2*x/pi))`

Amiet's TE noise model



- We previously found the far-field acoustic pressure of a blade in terms of the pressure difference at a given chordwise location on the airfoil:

$$\tilde{p}(\mathbf{x}, \omega) \approx \frac{-i\omega x_2 e^{ik_0 r_g - ik_0 M x_1}}{2c_\infty r_g^2} \int_{-c}^0 \Delta \tilde{p}(\mathbf{y}_1, k_3^{(o)}, \omega) e^{-ik_1^{(o)} y_1} dy_1$$

- Far-field spectral density of the acoustic pressure:

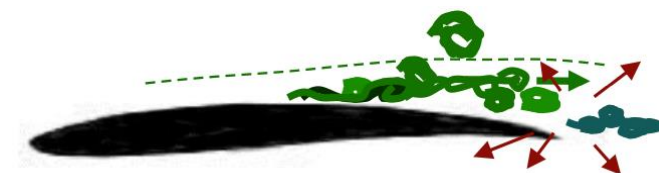
$$S_{pp}(\omega) = \frac{\pi}{T} E[\tilde{p}^*(\omega) \tilde{p}(\omega)] \quad \Rightarrow \quad S_{pp}(\mathbf{x}, \omega) \approx \left(\frac{\omega x_2}{2c_\infty r_g^2} \right)^2 R_\infty \int_{-c}^0 \int_{-c}^0 \underbrace{\frac{\pi}{TR_\infty} E[\Delta \tilde{p}^*(\mathbf{y}_1, k_3^{(o)}, \omega) \Delta \tilde{p}(\mathbf{y}'_1, k_3^{(o)}, \omega)]}_{= \frac{1}{\pi} \phi_{qq}(\mathbf{y}_1, \mathbf{y}'_1, k_3^{(o)}, \omega)} e^{-ik_1^{(o)}(\mathbf{y}'_1 - \mathbf{y}_1)} dy_1 dy'_1$$

ϕ_{qq} – spanwise wavenumber transform of the cross-spectrum of pressure difference fluctuations between any two points on the airfoil surface

$T = R_\infty / U_\infty$ – averaging time – the time for the turbulent volume to pass over the blade

$R_\infty = b/2$ – turbulent region size – taken as the wetted span of the blade

Amiet's TE noise model



$$S_{pp}(\mathbf{x}, \omega) \approx \left(\frac{kc_2}{2\pi r_g^2} \right)^2 \left(\frac{b}{2} \right) S_{pp}(0, \omega) l_{y_3}(\omega) \left| \mathcal{L}_{TE} \left(k_1^{(o)}, k_3^{(o)}, \omega, M \right) \right|^2$$

directivity coefficient $\times$ **Wall pressure spectrum near TE** $\times$ **Spanwise coherence length** $\times$ *Unsteady lift response function*

- c – airfoil chord
- b – span
- r_g – effective observer distance
- $U_\infty, \rho_\infty, c_\infty$ – freestream velocity, density, and speed of sound
- M – freestream Mach number, U_∞/c_∞
- ω – angular frequency [rad/sec]
- k – wavenumber [rad/m]
- ϕ_{22} – 2D (planar) wavenumber spectra of normal velocity fluctuations in x_1 and x_3
- $\mathcal{L}_{LE}$ – Unsteady lift transfer function

$$k = \frac{\omega}{c_\infty} \quad k_1 = \frac{\omega}{U_\infty}$$

$$k_o = \frac{\omega}{\beta^2 c_\infty} \quad \beta^2 = 1 - M^2$$

$$r_g = \sqrt{x_1^2 + \beta^2(x_2^2 + x_3^2)}$$

$$k_1^{(o)} = k_o \left(\frac{x_1}{r_g} - M \right)$$

$$k_3^{(o)} = \frac{k_o x_3 \beta^2}{r_g}$$